Comparative Interaction

Toward a Scientific Method for Comparing Computational Interactions

Formal research paper

Abstract

Comparisons of computational interactions are commonly reduced to task time, error rate, preference, usability scores, feature counts, or interface-level judgments. Such comparisons are scientifically incomplete because they often fail to establish that the interactions share a target, task, population, boundary, observation model, or admissible comparison relation. They also collapse distinct dimensions - semantics, interaction structure, opportunity, behavior, strategy, accessibility, discoverability, recovery, cost, learning, and measurement quality - into a single ranking.

This paper develops a cross-disciplinary method for comparing computational interactions independently of implementation. It integrates formal equivalence, process conformance, measurement theory, human factors, accessibility, multi-criteria analysis, and the preceding interaction-research framework. The central result is a gate-and-profile architecture. A comparison must first pass admissibility gates establishing declared targets, correspondence mappings, scope conditions, measurement comparability, and uncertainty. It then produces a typed comparison profile rather than an unqualified score. Aggregation is permitted only when a decision purpose, weighting rule, compensation assumptions, uncertainty treatment, and stakeholder authority are declared.

The paper concludes that computational interactions can be compared scientifically, but not by a universal metric. Scientific comparison requires a comparison contract, typed evidence, dimension-specific measures, explicit invariants, and rules for equivalence, dominance, trade-off, and incomparability. The reusable artifacts are a comparison ontology, taxonomy, contract, protocol, evaluation framework, scorecard, admissible-comparison rules, and falsification matrix.

1. Problem Statement

A computational interaction is not identical to its implementation, interface, trace, outcome, or user judgment. The same computational target may be projected through a graphical interface, command language, API, conversational system, or assistive modality. These projections may preserve operation semantics while changing exposed actions, signaling, path length, error recovery, auditability, motor demands, learning requirements, and strategy.

A scientific comparison must therefore answer two questions in order. First: are the candidate interactions comparable under a declared frame? Second: on which dimensions are they equivalent, different, better, worse, or incomparable? Most evaluation practice begins with the second question and leaves the first implicit.

The primary contribution of this paper is a method that separates comparison admissibility from comparative judgment. The method prevents three recurring errors: comparing unlike targets as though they were variants of one interaction; treating a metric as a complete construct; and converting multidimensional differences into a total ranking without an explicit decision model.

2. Foundations Inherited from the Research Program

The method inherits six constraints. Representation theory establishes that a projection selectively preserves, transforms, omits, and introduces properties. Opportunity theory separates possible, permitted, reachable, exposed, signaled, believed, selected, executed, successful, and reversible action relations. Interaction theory requires declared participants, boundaries, coupling, timescale, events, transitions, and completion. Behavior theory treats traces as evidence generated under an observation model rather than as transparent behavior. Measurement theory requires a declared measurand, operationalization, scale, uncertainty, validity domain, and permissible interpretation. Interaction-similarity theory distinguishes token identity, type identity, typed equivalence, graded similarity, and incomparability.

These constraints jointly imply that comparison is not a single function applied to two interfaces. It is an argument supported by mappings, measurements, and scope conditions.

3. Literature Foundations

3.1 Formal equivalence and conformance

Formal methods compare systems relative to observables and abstraction rules. Trace equivalence, simulation, bisimulation, testing equivalence, refinement, and observational equivalence preserve different properties. Process mining similarly separates fitness, precision, generalization, and simplicity when comparing event logs and process models. Alignment-based conformance identifies correspondences and deviations rather than assuming event-token identity. These traditions support a central rule: comparison relations must be typed before they are computed.

Conformance is not the same as superiority. A trace may conform closely to a model while imposing high human cost; a model may permit desired behavior while also permitting unsafe or irrelevant behavior. Formal relations therefore constrain comparison but do not exhaust it.

3.2 Measurement and context of use

Measurement theory requires stable representational relations, permissible transformations, and uncertainty statements. Metrological traceability concerns the documented chain linking results to references. In interaction research, the relevant chain links records to detected events, attributed behavior, derived variables, constructs, and claims.

ISO 9241-11 defines usability relative to specified users, goals, and contexts of use, with effectiveness, efficiency, and satisfaction as outcomes. This supports context-indexed comparison rather than universal interface quality. A result for one population, task, device, or environment does not automatically transport to another.

3.3 Accessibility and non-compensatory requirements

Accessibility standards show why some dimensions cannot be averaged away. WCAG separates perceivability, operability, understandability, and robustness into testable success criteria. Failure of keyboard access or text alternatives cannot be scientifically neutralized by faster performance for another population. Accessibility therefore motivates threshold, veto, and subgroup-preservation rules rather than purely compensatory scoring.

3.4 Multi-criteria decision analysis

Multi-criteria decision analysis distinguishes description from preference aggregation. Weighted sums assume commensurability and compensation: a deficit on one criterion can be offset by strength on another. Outranking methods permit partial orders and incomparability when evidence, thresholds, or stakeholder priorities do not justify a total ranking. Comparative interaction should adopt this discipline. A scientific profile describes differences; a decision model may rank alternatives, but the ranking is conditional on declared values and constraints.

4. Comparison Ontology

Concept	Definition
Comparison object	The interaction token, type, projection, trace distribution, opportunity profile, strategy, or system pair being compared.
Reference frame	The declared target, task, population, context, boundary, timescale, and observation model.
Correspondence mapping	A mapping between entities, states, operations, events,

Concept	Definition
	outcomes, or measurements across candidates.
Invariant	A property required to remain unchanged for the comparison claim.
Dimension	A typed aspect of comparison such as semantic, structural, opportunity, behavioral, accessibility, or cost.
Measure	An operational rule producing evidence about a dimension.
Tolerance	A declared interval or threshold within which differences count as equivalent or acceptable.
Uncertainty	Incomplete knowledge attached to records, mappings, estimates, model assumptions, or transport.
Admissible transformation	A change allowed without invalidating the comparison, such as renaming, modality substitution, or temporal rescaling.
Veto condition	A non-compensatory failure that blocks equivalence, dominance, or deployment under the declared purpose.
Comparison result	A typed relation: identical, equivalent, similar, dominant, trade-off, or incomparable.
Decision rule	An optional value-laden rule for ranking or selecting candidates from the scientific profile.

5. Interaction Comparison Taxonomy

Dimension	Comparison question	Primary evidence
Semantic preservation	Whether operations, state meanings, constraints, and outcome interpretations correspond.	Formal specification, model mapping, expert validation.
Interaction preservation	Whether participant roles, coupling, protocol, boundaries, transitions, and completion conditions correspond.	Interaction Contracts; protocol and transition analysis.
Opportunity preservation	Whether feasible, permitted, reachable, exposed, signaled, believed, and recoverable actions correspond.	Opportunity matrices, discovery studies, authorization analysis.
Behavioral preservation	Whether attributed event and episode distributions correspond under compatible observation models.	Trace alignment, sequence models, distributional tests.
Strategy preservation	Whether inferred conditional organization, information use, switching, and recovery policies correspond.	Multi-task evidence, perturbation tests, strategy models.
Accessibility	Whether relevant populations can perceive, operate, understand, and robustly use the interaction.	Standards conformance plus user evidence.
Discoverability	Whether operations and relations can be found under declared prior knowledge and search conditions.	Unaided discovery, time-to-discovery, cue recognition.
Reversibility and recovery	Whether consequences can be undone, repaired, compensated, or safely explored.	Recovery graph, repair success, residual harm and cost.
Interaction cost	Resources consumed during performance.	Time, actions, motor distance, attention, memory, energy, money, risk.
Learning cost	Resources and exposure required to achieve a declared criterion.	Learning curves, trials-to-criterion, retention and transfer.
Measurement quality	Whether evidence is valid, reliable, comparable, traceable, and uncertainty-bounded.	Measurement contracts, invariance tests, uncertainty budgets.

6. Admissibility Gates

A comparison should not proceed directly to metrics. It must pass the following gates. Failure does not imply that the candidates cannot be discussed; it means the proposed scientific comparison is unsupported or must be narrowed.

Gate	Question	Required evidence	Typical failure
G1 Target	Is the underlying target, function, or declared purpose sufficiently aligned?	Target definitions; semantic correspondence map.	Different goals are treated as variants of one task.
G2 Boundary	Are participants, episode boundaries, timescales, initial conditions, and completion rules aligned?	Interaction Contracts.	One candidate includes setup or recovery while the other excludes it.
G3 Opportunity	Are differences in permissions, capabilities, backend functions, and reachable outcomes identified?	Opportunity Contracts.	Interface differences are credited for backend capability differences.
G4 Observation	Do records, event detection, segmentation, attribution, and missingness permit comparison?	Behavioral Evidence Contracts.	Different logging coverage is interpreted as behavioral difference.
G5 Measurement	Are constructs, scales, instruments, populations, and uncertainty compatible?	Measurement Contracts; invariance and reliability evidence.	Scores from non-equivalent instruments are directly compared.
G6 Correspondence	Can entities, actions, events, outcomes, and costs be defensibly mapped?	Mapping table with ambiguous and unmapped elements.	Labels are assumed to identify equivalent operations.
G7 Ethics and accessibility	Are safety, rights, exclusion, and accessibility constraints represented?	Veto rules; subgroup analysis; standards evidence.	Average gains conceal a blocked or harmed subgroup.

7. Comparison Relations

After admissibility, a result should use one of six relations rather than the undifferentiated term “better.”

Relation	Meaning	Evidence
Token identity	The same historically bounded episode or record lineage.	Stable identity and provenance.
Typed equivalence	All required invariants fall within declared tolerances for a named dimension.	Equivalence tests or formal proof.
Similarity	A graded relation under a declared metric and comparison frame.	Distance, distribution, or model-based estimate with uncertainty.
Dominance	One candidate is no worse on all mandatory dimensions and better on at least one.	Comparable dimensions and directionality.
Trade-off	Candidates differ in opposing directions and no value-neutral ordering follows.	Profile plus optional stakeholder decision model.
Incomparability	No defensible common frame, mapping, measurement, or preference relation exists.	Documented gate failure or unresolved uncertainty.

8. Comparative Evaluation Framework

The recommended evaluation output has three layers.

Layer 1 - invariants and vetoes. These establish semantic target, safety, authority, accessibility minima, and other non-negotiable constraints. A candidate failing a veto is not equivalent or admissible for the declared use, regardless of aggregate performance.

Layer 2 - typed profile. Each dimension is reported with operational definition, estimate, uncertainty, evidence quality, and relevant subgroup or context. This is the scientific result.

Layer 3 - optional decision synthesis. Ranking or selection may use dominance, lexicographic rules, constrained optimization, utility functions, or outranking. The decision rule must be separated from the scientific profile and must disclose weights, thresholds, compensation assumptions, and authority to set them.

9. Interaction Scorecard

Domain	Indicators	Result form	Required note
Target and semantics	Operation and outcome preservation	Pass / partial / fail	Formal mapping; unresolved mismatches
Interaction structure	Roles, coupling, protocol, completion	Profile	Structural and temporal deviations
Opportunity	Executable, exposed, signaled, believed, recoverable sets	Profile	False-positive and hidden opportunity rates
Behavior	Trace and outcome distributions	Profile	Compatible segmentation and uncertainty required
Strategy	Conditional organization and recovery	Profile / latent	Requires convergent evidence across contingencies
Accessibility	Perceivable, operable, understandable, robust	Threshold + profile	Report by relevant access needs
Discoverability	Unaided discovery and signaling effectiveness	Profile	Separate exposure from belief
Reversibility	Undo, repair, compensation, residual effects	Threshold + profile	Irreversible harms are veto candidates
Performance cost	Time, action, motor, cognitive, financial, risk	Profile	No universal common unit
Learning cost	Exposure, instruction, trials, retention, transfer	Profile	Separate familiarity and transfer
Measurement quality	Validity, reliability, invariance, missingness, uncertainty	Quality grade	Low quality constrains all downstream claims

10. Comparison Protocol

- 1. State the decision-independent scientific question and identify the candidate interactions.
- 2. Declare target semantics, task, users or agents, context, initial-state distribution, episode boundary, timescale, and success conditions.
- 3. Instantiate an Interaction Contract, Opportunity Contract, Behavioral Evidence Contract, and Measurement Contract for each candidate.
- 4. Construct correspondence mappings for entities, states, operations, events, outcomes, strategies, and measures. Mark one-to-many, ambiguous, and unmapped elements.
- 5. Declare invariants, admissible transformations, equivalence tolerances, mandatory thresholds, and veto conditions before examining comparative outcomes.
- 6. Test admissibility gates. Narrow the claim or declare incomparability where gates fail.
- 7. Select dimension-specific measures and justify their scales, directionality, uncertainty, and subgroup relevance.
- 8. Collect or derive evidence using compatible observation and attribution rules. Preserve raw records and provenance.
- 9. Estimate typed differences with uncertainty. Use equivalence tests for sameness claims and superiority tests only for directional claims.
- 10. Report the multidimensional profile, including null, conflicting, and subgroup-specific results.
- 11. Apply an optional decision rule only after the profile is complete. Conduct weight, threshold, and uncertainty sensitivity analysis.
- 12. State validity domain, transport limits, unresolved mappings, alternative explanations, and falsifying evidence.

11. Admissible Comparison Rules

- No comparison without a declared comparison object and reference frame.
- No claim of semantic preservation from equal task completion alone.

- No claim of behavioral preservation from identical aggregate outcomes alone.
- No comparison of traces without compatible observation, segmentation, and attribution models.
- No inference of strategy identity from one trace or one successful sequence.
- No aggregation across dimensions without a declared decision purpose and compensation rule.
- No averaging across populations when a mandatory subgroup threshold is relevant.
- No accessibility credit from technical conformance alone when actual operability is disputed; standards and user evidence play different roles.
- No cost comparison without declaring included resources, excluded externalities, and time horizon.
- No equivalence claim without tolerances and adequate precision to support them.
- No ranking when uncertainty intervals, missing mappings, or value conflicts support only a partial order.
- Implementation independence permits different realizations, not the omission of implementation-mediated constraints that alter actual interaction.

12. Comparative Cases

12.1 GUI and command line over the same operation set

A GUI and CLI may preserve backend operations and outcomes while differing in exposure, signaling, composition, auditability, motor demands, memory demands, and learning cost. Semantic equivalence can coexist with opportunity and behavioral inequivalence. Experts may obtain lower execution cost from the CLI while novices achieve higher discovery from the GUI. The result is typically a trade-off profile, not a universal winner.

12.2 API and conversational interface

An API may expose a precisely typed operation set with explicit errors, while a conversational interface paraphrases capabilities and may over-signal unsupported actions. Equal successful outcomes on benchmark tasks do not establish equivalent permission transparency, predictability, recovery, or auditability. Unsupported conversational claims create false-positive believed opportunities and should be represented as a veto or severe mismatch where consequential actions are involved.

12.3 Accessibility variants

Visual, auditory, tactile, and multimodal projections may preserve target semantics while changing sequence, latency, spatial comparability, memory burden, error detection, and recovery. Comparison requires population- and assistive-technology-specific evidence. Aggregate performance across users can conceal exclusion. The correct output is a subgroup-indexed accessibility and opportunity profile.

12.4 Novice and expert use

The implementation and exposed operations may be identical while believed actions, strategy, search, confidence, and recovery differ. This is not an interface-only comparison. The reference frame must include prior experience, and learning cost should be compared through longitudinal or transfer-sensitive designs rather than single-session performance alone.

12.5 Simulated and observed interaction

A simulation may match transition frequencies or task outcomes yet fail to preserve causal mechanisms, adaptation, social coordination, or measurement processes. Model-to-observation comparison must separately report calibration, trace conformance, causal adequacy, and transport. Simulation should not be treated as a substitute population without validation.

13. Statistical and Formal Analysis

Claims of equivalence should use equivalence or non-inferiority designs with prespecified smallest effects of interest, not failure to reject a difference. Repeated measures, hierarchical populations, learning, and task heterogeneity require multilevel models. Time-to-discovery and abandonment are event-time outcomes; errors and recovery are often recurrent events; traces may require sequence alignment, process conformance, or probabilistic model comparison.

Multiple dimensions create multiplicity and selective-reporting risks. Confirmatory dimensions should be preregistered. Exploratory profiles should remain labeled as exploratory. Missingness is often informative: an unobserved action may reflect logging limits, access barriers, or abandonment. Uncertainty should propagate from record detection and correspondence mappings through construct estimates and final comparisons.

Formal equivalence can establish strong guarantees within an abstraction, but empirical claims require validating that the abstraction preserves the phenomena of interest. Conversely, empirical similarity does not establish formal equivalence or causal identity.

14. Decision Synthesis without Conceptual Collapse

A scorecard may support decisions, but a universal interaction score is rejected. Aggregation is appropriate only when criteria are commensurable or a defensible value function is supplied. Weighted sums permit compensation and can hide threshold failures. Lexicographic rules prioritize one criterion before others. Constrained optimization maximizes selected objectives subject to non-negotiable limits. Outranking supports partial orders where one candidate is sufficiently strong on important criteria without being vetoed by severe weaknesses.

Sensitivity analysis is mandatory. A decision is robust only if reasonable variation in weights, thresholds, measurement uncertainty, and population assumptions does not reverse it. When rankings are unstable, the correct result is conditional preference or incomparability, not false precision.

15. Falsification Matrix

Claim challenged	Falsifying evidence	Consequence
Typed profiles add no value	A validated scalar measure predicts all relevant outcomes and decisions across systems, populations, tasks, and contexts without hidden trade-offs.	Replace the profile with that measure within its proven domain.
Admissibility gates are unnecessary	Comparisons remain valid when targets, boundaries, observation models, or mappings differ materially.	Remove unsupported gates.
Implementation independence is incoherent	Every meaningful comparison depends irreducibly on implementation-specific details.	Narrow the framework to implementation classes.
Opportunity comparison is redundant	Semantic and behavioral measures fully explain exposure, belief, permission, and recovery differences.	Remove opportunity as a separate dimension.
Strategy comparison is unidentifiable	No protocol distinguishes strategy from trace variation or task response with acceptable reliability.	Treat strategy as exploratory only.
Accessibility can be fully compensatory	Empirical and normative evidence supports averaging access failures against gains elsewhere for the declared use.	Remove veto treatment for that domain.
Comparison contracts do not improve reproducibility	Independent teams using the contract do not converge more than teams using ordinary reports.	Revise or reject the contract.
Profiles cannot support decisions	Decision makers cannot use typed profiles more reliably than existing evaluations.	Reduce the framework to descriptive science.

16. Threats to Validity

The framework is broad and may appear to solve comparison by requiring extensive declarations. Its value therefore depends on demonstrating improved reproducibility, error detection, and decision quality relative to simpler evaluation reports. Some comparisons will remain underdetermined because goals, values, or correspondence mappings are contested. This is not necessarily methodological failure; incomparability may be the scientifically correct result.

Construct proliferation is another risk. The taxonomy should be reduced when dimensions are empirically redundant under a declared domain. Conversely, low correlations do not automatically prove conceptual independence because instruments may be unreliable or contexts heterogeneous.

The framework also risks privileging interactions with enumerable tasks and state spaces. Open-ended creative, social, and institutional interactions may require interpretive or qualitative comparisons alongside formal and quantitative evidence. Such evidence remains admissible when its observation, coding, reflexivity, and inference rules are explicit.

Finally, implementation independence has limits. Rendering latency, hardware, modality, network behavior, and device constraints can alter coupling, opportunity, and cost. The theory compares interactions independently of any particular implementation language or product, but it cannot discard implementation properties that causally alter the phenomenon.

17. Conclusion

Computational interactions can be compared scientifically when comparison is treated as a structured evidential argument rather than a feature checklist or universal score. The minimum method is a gate-and-profile architecture.

First, admissibility gates establish a shared target, boundary, opportunity structure, observation model, measurement system, correspondence mapping, and ethical or accessibility constraints. Second, a typed profile compares semantic preservation, interaction structure, opportunity, behavior, strategy, accessibility, discoverability, reversibility, performance cost, learning cost, and measurement quality. Third, an optional decision model may aggregate or rank only after its values, thresholds, compensation assumptions, and uncertainty are declared.

The central conclusion is therefore conditional but positive: a general scientific method for comparative interaction is justified. It is not a universal metric and does not imply that every pair of interactions is comparable. Its scientific strength lies in making equivalence, difference, trade-off, dominance, and incomparability explicit and reproducible.

18. Reusable Artifact A - Comparison Contract

Field	Required declaration
Identity	Candidate interactions; versions; provenance; token or type comparison.
Reference frame	Target, task, users/agents, context, initial state, boundary, timescale, completion.
Correspondence	Entities, states, actions, events, outcomes, strategies, measures; ambiguous and unmapped items.
Invariants	Properties required to remain fixed by dimension.
Transformations	Renaming, reordering, abstraction, modality substitution, temporal rescaling, implementation substitution.
Dimensions	Selected comparison dimensions and reasons for inclusion or exclusion.
Measures	Measurands, instruments, scales, estimators, directionality,

Field	Required declaration
	reliability, validity.
Thresholds	Equivalence bounds, minimum acceptable values, smallest effects of interest, vetoes.
Uncertainty	Record, mapping, sampling, model, instrument, and transport uncertainty.
Evidence	Formal proofs, records, experiments, qualitative evidence, standards conformance, expert review.
Decision synthesis	Optional weights, priorities, compensation, authority, sensitivity analysis.
Scope	Population, task, projection, context, time horizon, and invalidating conditions.

19. Reusable Artifact B - Admissible Transformation Matrix

Transformation	Admissible?	Conditions	Affected dimensions
Renaming or identifier substitution	Usually	Preserve referents and operation semantics.	Semantic, structural
Visual restyling	Sometimes	No change to signaling, legibility, salience, or access.	Semantic; possibly opportunity
Modality substitution	No by default	Requires population-specific access and sequence evidence.	Semantic may hold; behavior/access may differ
Action reordering	Sometimes	Preserve causal dependencies and outcomes.	Protocol, trace, cost
Temporal rescaling	Sometimes	Preserve ordering and relevant timing relations.	Structural but not necessarily performance
Aggregation or abstraction	Sometimes	Declare omitted events and preserved observables.	Type-level, not token-level
Implementation substitution	Sometimes	Preserve externally relevant semantics, latency, failure, authority, and state.	Semantic/formal; empirical revalidation needed
Automation of user steps	No by default	Changes participant roles, coupling, opportunity, and strategy.	Interaction structure and behavior
Permission or policy change	No	Changes admissible action structure.	Opportunity and semantics
Training intervention	No for behavioral equivalence	Changes agent state, belief, strategy, and learning cost.	Population/context comparison required

20. Reusable Artifact C - Minimal Reporting Template

- Comparison claim: exact wording and typed relation.
- Candidates and provenance.
- Reference frame and validity domain.
- Contracts used and gate results.
- Correspondence mapping and unresolved items.
- Invariants, transformations, tolerances, and vetoes.
- Dimension-by-dimension measures, estimates, uncertainty, and evidence quality.
- Subgroup and context variation.
- Alternative explanations and falsifying evidence.
- Optional decision rule, sensitivity analysis, and conditional recommendation.

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